

Ziyi Sun

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College of Engineering, Ocean University of China, Qingdao 266100, China

EDUCATION

- **Institute of Automation, Chinese Academy of Sciences** Sep. 2027 – Jun. 2030 (Expected)
M.Eng. in Computer Application Technology Beijing, China
- **Ocean University of China (985 / Double First-Class)** Sep. 2023 – Jun. 2027 (Expected)
B.Eng. in Mechanical Design, Manufacturing and Automation Qingdao, China
 - GPA: **3.7/4.0** Rank: **1/71** Average: 89.2
 - Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

RESEARCH EXPERIENCE

- **Roboparty Lab** Jul. 2026 – Oct. 2026
Frontier Research and Open Source Intern (Mentor: Jagger) Beijing, China
 - Contribute to frontier research and open-source development for embodied intelligence and robot learning systems.

PUBLICATIONS & PATENTS

C=CONFERENCE, S=IN SUBMISSION, P=PATENT, SW=SOFTWARE

[C.1] A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot.

Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng*.

IEEE International Conference on Real-time Computing and Robotics (IEEE RCAR), 2026. **Best Paper Finalist.**

- Proposed a two-stage framework combining multimodal perception, expert demonstration, PPO, AHTA-CPG, and domain randomization.
- Achieved 70%–90% success over 60 real water-tank trials.

[C.2] Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator.

Dingyang Yu, Yuanrong Zhong, Qi Chen, Siming Ke, Qianbei Luo, **Ziyi Sun**, Xunqi Zhao.

IEEE International Conference on Computer, Control and Robotics (ICCCR), 2025.

- Presented the mechanical design and control framework of a biomimetic robotic eel with an omnidirectional quadruple wire-driven body and CPG-based locomotion control.

[S.1] CHOREO: Every Humanoid Skill as a Traject.

Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.

Under review at AAAI 2027 (CCF-A).

- Unified SkillMotion representation and Tracker interface with boundary compatibility assessment, trajectory fusion, and bridging-action retrieval for reusable multi-source humanoid skills.

[S.2] A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.

Ziyi Sun, Changhong He, et al.

Under review at *Journal of Marine Science and Engineering (JMSE)*.

- Developed a unified simulation and benchmarking pipeline for scalable robot learning, integrating CAD-based robot models and reproducible comparisons of centralized vs. distributed multi-AUV nodule collection.

[P.1] A High-Stability PTO System for Duck-Type Wave Energy Converters.

Zhi Han, **Ziyi Sun**, Haoran Chen, Hailong Zhao, Wenchi Wei, Xueyao Wang, Hongda Shi.

Invention Patent (First Student Inventor).

[SW.1] Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0.

Haoyuan Cheng, Chuanyun Su, Zhichao Yan, Shuning Wang, **Ziyi Sun**.

Software Copyright (No. 2024SR0140235).

PROJECTS

- **Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection** Dec. 2025 – Dec. 2026
Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing OUC SRDP
 - Designed a biomimetic robotic fish with continuously adjustable body length (1.2–1.5 m) for deep-sea cage cruise and narrow-space inspection.
 - Fused school-fish visual tracking with water-quality sensing for joint monitoring and early warning of aquaculture risks.
- **Cross-Domain Wire-Driven Biomimetic Eel Robot** Dec. 2024 – Nov. 2025
Role: Project Leader | Tools: vision tracking, Gaussian Splatting, damage detection Provincial Innovation Program
 - Developed an eel-like robot with cruising and active depth control for floating offshore wind dynamic-cable inspection.
 - Integrated visual tracking, 3D reconstruction, and damage detection for cable modeling, defect localization, and risk warning.
- **Modular Adaptive Deep-Sea Resource Collection and Locomotion Platform** Dec. 2024 – Nov. 2025
Role: Core Member | Tools: Fluent, mechanical design Research Project
 - Designed a modular adaptive deep-sea mining vehicle for complex seabed terrain, including Fluent-based plume analysis during locomotion/mining and plume-suppression structures.
 - Combined an active four-track chassis with a compliant collection structure to improve geological adaptability on the seabed.
- **High-Stability Duck-Type Wave Energy Converter with Pendulum PTO** Dec. 2024 – Nov. 2025
Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks National Innovation Program
 - Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and a combined PTO with nonlinear adaptive regulation.

SKILLS

- **Robot Platform:** Unitree G1
- **Programming & ML:** Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:** SolidWorks, AutoCAD
- **Languages:** Chinese (Native); English (CET-4: 591, CET-6: 482)

HONORS AND AWARDS

- **Shandong Provincial Government Scholarship** 2025
- **First-Class Comprehensive Scholarship, Ocean University of China** 2024 & 2025
- **Outstanding Student / Outstanding Student Cadre, Ocean University of China** 2024 & 2025
- **National First Prize, National 3D Digital Innovation Design Competition** 2025 & 2026
- **National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)** 2025
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** 2025
- **National Third Prize, 18th National College Student Energy Conservation Competition** 2024
- **National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest** 2024
- **National Second Prize, 14th Marine Vehicle Design and Production Competition** 2025
- **National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition** 2026
- **Science and Technology Innovation Pioneer Team, College of Engineering, OUC** 2024

LEADERSHIP EXPERIENCE

- **Ocean University of China** 2023 – Present
Class Representative / Class Monitor Qingdao, China
- **College of Engineering Debate Team, OUC** 2024 – 2025
Team Leader Qingdao, China
- **Student Association for the Study of Socialism with Chinese Characteristics, OUC** 2024 – 2025
Director of Event Organization Qingdao, China

VOLUNTEER EXPERIENCE

- **Volunteer, Qingdao Marathon** 2023
- **Member, National College Student Volunteer Outreach Team** 2024
- **Outstanding Individual in Summer Social Practice, OUC** 2024